

Site Analysis

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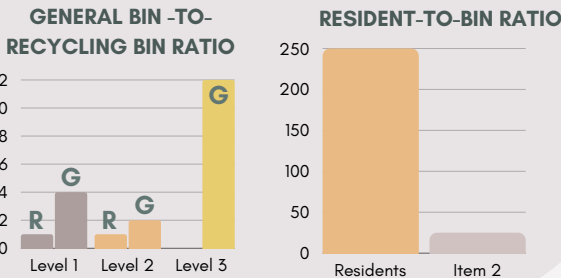
Problem Statement

How might we create an effective cleaning solution to improve hygiene standards and efficiency for cleaning staff?

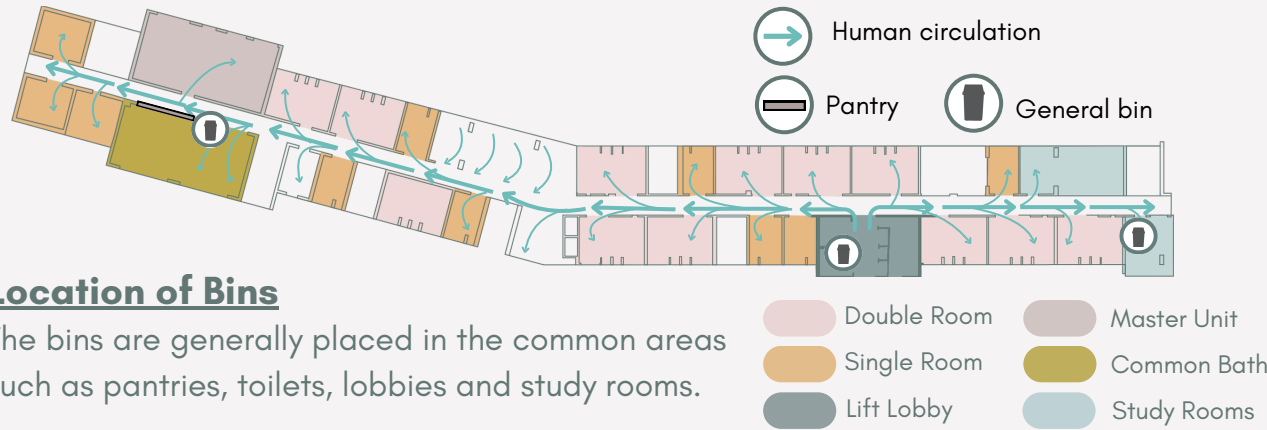
Cleaner's Routine



BINS ANALYSIS



HOSTEL PLAN



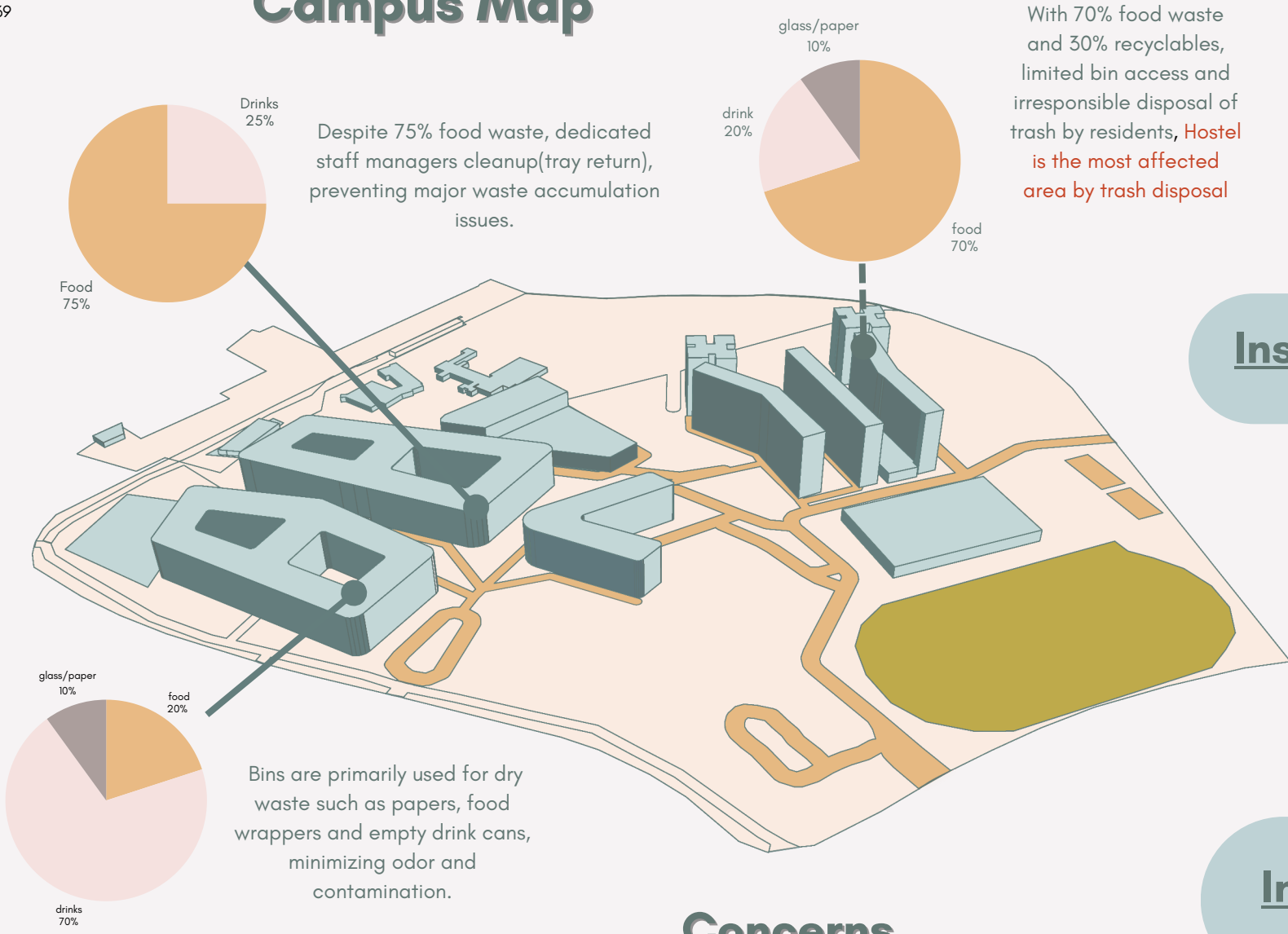
Location of Bins

The bins are generally placed in the common areas such as pantries, toilets, lobbies and study rooms.

Odour analysis

Odour that flows through the hostel is mainly sourced from the bins near the common areas which have denser circulation.

Campus Map



Concerns

1. Accessibility

- Recycling:** Limited bins (only on lower floors) discourage recycling and are inaccessible
- Waste bins:** Insufficient bins per floor create inconvenience, especially for those with mobility issues.
- Chutes:** Lack of awareness about rubbish chute availability.

3. Odor

- Smell:** Overflowing bins = bad smells.
- Decomposition:** Food waste decomposes, worsening smells.
- Collection:** Infrequent emptying creates the problem.(especially on weekends)
- Pests:** Odors attract pests.
- Bacteria:** Promotes bacterial/mold growth.
- Environment:** Bad smells negatively impact living environment.

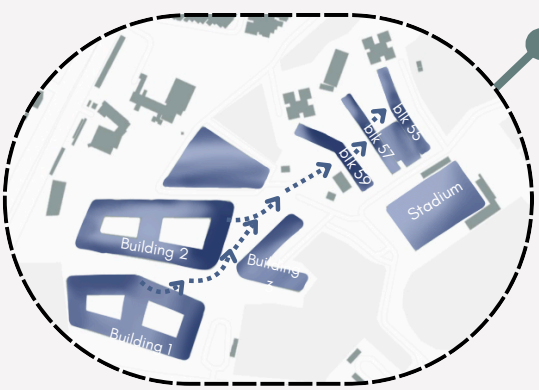
2. Overcrowding

- Obstacles:** Overflowing bins obstruct pathways.
- Hygiene:** Full bins lead to improper disposal and spills.
- Leakage:** Food waste leaks from bins.
- Cleaning:** Overflows increase cleaning burden.

4. Irresponsible behavior of users

- Contamination:** Food/liquid waste thrown into recycling bins.
- Segregation:** Residents disregard waste segregation despite labeling.
- Convenience:** Trash is disposed of in the nearest bin, regardless of type.
- Responsibility:** Minimal effort to improve waste disposal habits.

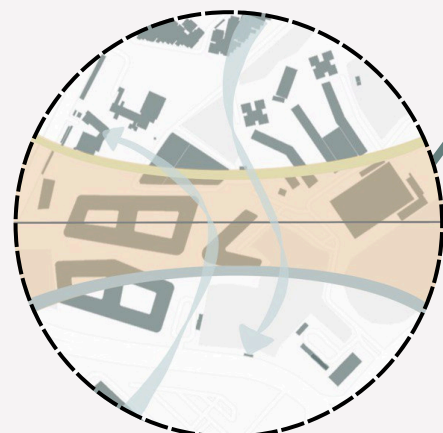
Foot Traffic Analysis



Insights

Bins near canteens and exits from academic buildings may fill up faster due to more personal waste like takeaway containers, plastic bottles, or general trash from students bringing food back.

Environmental Analysis



Insights

- East-to-west sunlight heats bins near hostels, accelerating waste decomposition, odor, and bacterial growth.
- Due to a large surface area of buildings exposed to the wind, foul smells from overflowing or poorly maintained bins may travel to living spaces, classrooms and other gathering areas

DESIGN GOALS

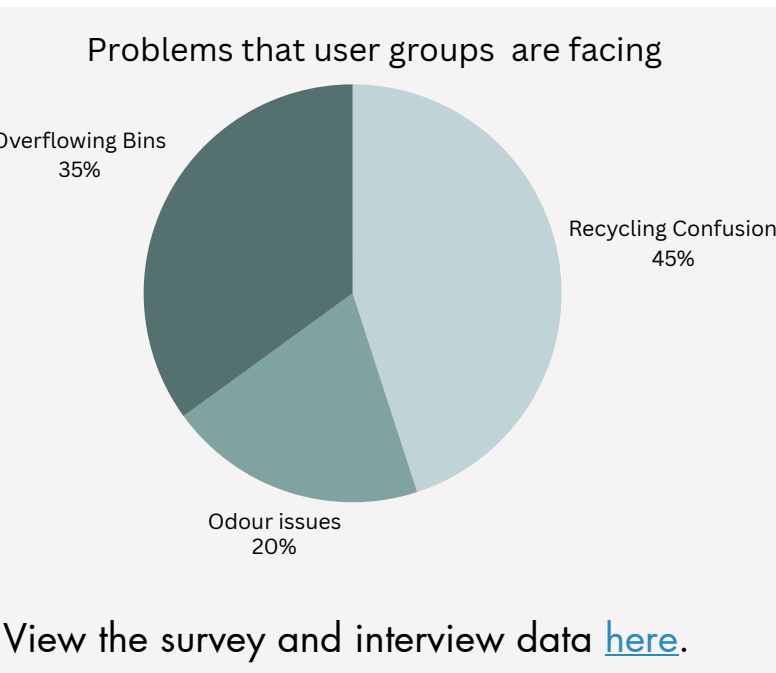
The hostel, located on a major trash route, suffers from inadequate waste management. **Overcrowded** and insufficient recycling bins create unsanitary conditions, forcing constant bin checks and **manual sorting** for the cleaning staff. Residents' misuse of recycling bins further hinders proper disposal.

Our solution aims to:

- Reduce overflowing bins by increasing recycling capacity
- Prevent improper waste disposal through clear segregation.
- Minimize odor
- Ensure waterproof and leakproof containment.
- Promote sustainability.

Problem Statement

How might we create an effective cleaning solution to improve hygiene standards and efficiency for cleaning staff?



Pain Points

Smelly Bins: Trash bins have unpleasant odors.
Recycling Confusion: Users are unsure which bin to use.
Overflowing Bins: Bins frequently overflow.
Mixed Recycling: Recyclables are often mixed with trash.

Legend

- A** Activity
- E** Environment
- I** Interaction
- O** Object
- U** User

User Group

- Primary User
- Cleaners
- Secondary User
- Professor
 - Admin Staff
 - Fablab Staff
 - Student

Uncle Ben

Age : 65
Gender : Male
Cleaner



7.00 am

Arrived from Malaysia, starts cleaning corridors

😊 Neutral



9.00 am

Clear the trash from trash bins

😞 Disgusted



2.00 pm

Clear recycling bins since students **misplace trash**

😡 Annoyed



4.00 pm

Clean bins to prevent **smell** from spilled drinks

😫 Exhausted



7.00 pm

Putting trash into the floaters to finish of work

😫 Exhausted

Ms. Jenna

Age : 32
Gender : Female
Fablab staff



7.00 am

Arrived in SUTD, meet other staffs

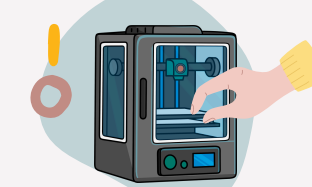
😊 Neutral



8.00 am

Assisting students with issues in other facilities

😄 Happy



10.00 am

Printing maintenance, frustrated by filament waste.

😡 Annoyed



8.00 pm

Finishing project, but students throwing **recyclable material in general trash**.

😡 Annoyed

Prof. Richard

Age : 45
Gender : Male
Professor



7:00am

Wakes up, leaves for university.

😊 Fresh



8:30am

Grabs breakfast on the way.

😄 Happy



8:55 AM

Becomes **confused about which recycling bin to use** for the trash.

😞 Confused



9:00am

Ends up throwing trash in the **overflowing dustbin**

😞 Disgusted



9:10am

Shows up to class 10 mins late

😡 Annoyed

Student Emma

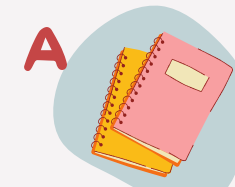
Age : 20
Gender : Female
Student



8:00 am

Wakes up and has a brief Yoga session

😊 Fresh



9:00 am

Arrives for DTI class.

😊 Neutral



12:00 pm

Goes to Canteen with friends for Lunch.

😄 Happy



12:30 pm

Returns the tray to the tray station. Sees the **spilled trash and overflowing bin**.

😡 Annoyed



5:00 pm

Goes to fifth-row activity.

😫 Exhausted